

In Pursuit of a PhD

Getting to, and surviving in graduate school.

Part I: Getting in...

Before Grad School

- Three things you need before you leave UMD
 - Course background
 - Base skill set
 - GRE's
- This is true whether you're an undergrad or a master's student!



Relevant Coursework

(what they expect you to know)

- “The big four”
 - Classical mechanics
 - Statistical Mechanics
 - Electrodynamics
 - Quantum Mechanics
 - ...more on these later!
- You probably won't have everything they expect.
- Shorten this gap by challenging yourself!
 - Undergrads: take cross listed, and external courses
 - Grads: challenging courses (fluids!)



That's a lot to know!
(but, there's more)

Base Skill Set

(the things they won't put on their list of requirements)

- Three big umbrellas
(because you'll all be good at *physics* when you're done!)
 - Math skills
 - Computational skills
 - Research experience
- The “intangibles” of graduate study
- Math skills can be enriched through out of department (and higher level) courses
 - Linear algebra
 - Complex analysis
 - Tensor analysis
 - Advanced Methods

Base Skill Set

(sorry, no informative comment here)

- Computational skills are important!
 - Mathematica/Maple
 - “Traditional” programming
- Your preference, but...
 - Mathematica, FORTRAN, C
- Research (at the graduate *or* undergraduate level)
 - All three of us did it at both levels
 - And it’s helped us immensely!

Research at UMD

- Josh
 - Loop Quantum Cosmology, Relativity theory
- Rob
 - Pion photoproduction, LQC, Gravity waves
- Subir
 - LQC, Yang-Mills Gravity
- Options!

Questions? Tough!

(there's one more *very* important thing)



The GRE's

GRE's

- The GRE's are more important than many of us would like to admit.
- Used as a “sifter” during admission
- A good score will almost always land you in a good school
- 600-700 ok, 700-800 better, but 800+ great! Aim high, and *prepare early!*

Part II: You made it!

(now how do you survive?)

Coursework in Grad School

- Before research begins, you do have to take some courses. The typical requirements vary by school.
- For example, at Columbia University its ~ 7 .
- We can split coursework into two parts:
 - Basic courses
 - Advanced, research specific courses

Basic Courses

- Classical Mechanics at the level of Goldstein
- E&M (2 semester sequence) at the level of Jackson
- Advanced Math Methods at the level of Hassani
- Quantum Mechanics (2 semester sequence) at the level of Gottfried, Merzbacher etc.
- Statistical Mechanics at the level of Pathria,

Advanced Courses

- These are field specific.
- Physics can be broadly divided into 4 different categories:
 - High Energy Physics (Expt. & Theory)
 - Astrophysics (Observation & Theory)
 - Condensed Matter Physics (Expt. & Theory)
 - Atomic, Molecular & Optical Physics

High Energy Physics

- Quantum Field Theory 1, 2 & 3
- Particle Physics 1 & 2
- General Relativity 1 & 2
- Super symmetry, Super gravity
- String Theory, Quantum Gravity
- Conformal Field Theory
- Scientific Computing (Experimental guys)

Astrophysics

- General Relativity 1 & 2 (optional)
- Advanced Statistical Mechanics
- Astrophysics 1 & 2
- High Energy Astrophysics
- Scientific Computing (Observational guys)
- Etc.

Condensed Matter Physics

- Advanced Statistical Mechanics
- Condensed Matter Physics I & II
- Quantum Field Theory I & II
- Scientific Computing (computational guys)

AMO

- Atomic Physics 1 & 2
- Optical Physics 1 or 2
- Condensed Matter 1 or 2
- Quantum Computing
- Biophysics
- Other exotic areas

PhD QUALIFYING EXAMS



JORGE CHAM ©THE STANFORD DAILY

Quals

- Qualifiers are a set of exams that you give in the first 1-1.5 years of your graduate school.
- The idea is to test you in senior level undergrad to beginning graduate level physics
- Infamous at institutions like UC Berkeley to “screen out” students.
- But that has changed.
- The right attitude to is consider them as something you do for yourself – to see if you learn basic level physics before you begin

Typical Exams and Pattern

- 6 exams over 3 days (2 hour exams each) followed by orals with a few faculty
 - Classical Mechanics
 - E&M
 - Quantum
 - Advanced Quantum & Relativity
 - General Physics I (really Stat Mech, Fluids, Optics etc)
 - General Physics 2 (HEP, Particle Physics, Expt. Methods)

Research

- Usually starts in 1st summer.
- For Theory, usually after 2nd year (HEP)
- Finding your area of interest.
- Getting an advisor
- Picking a problem
- 4 years later – Graduation!!!

And one last thing...

...a week in the life of a graduate student!

Today

◀

Day

Week

Month

▶

Q

CALENDARS

☐ Home
 ☒ GWU
 ☒ Homework
 ☒ Eating
 ☒ TAing
 ☒ Sleep

2009

all-day

1 AM

2 AM

3 AM

4 AM

5 AM

6 AM

7 AM

8 AM

9 AM

10 AM

11 AM

Noon

1 PM

2 PM

3 PM

4 PM

5 PM

6 PM

7 PM

8 PM

9 PM

10 PM

11 PM

Sunday, Nov 8

Monday, Nov 9

Tuesday, Nov 10

Wednesday, Nov 11

Thursday, Nov 12

Friday, Nov 13

Saturday, Nov 14

◀

November 2009

▶

Sun

Mon

Tue

Wed

Thu

Fri

Sat

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

1

2

3

4

5

6

7

8

9

10

11

12

+

📅

✉

📌

Time	Sunday, Nov 8	Monday, Nov 9	Tuesday, Nov 10	Wednesday, Nov 11	Thursday, Nov 12	Friday, Nov 13	Saturday, Nov 14
12:00 AM	Sleep	Sleep	Homework	Sleep	Homework	Homework	Sleep
1 AM							
2 AM							
3 AM							
4 AM							
5 AM							
6 AM							
7 AM							
8 AM							
9 AM							
10 AM						8:30 AM Computational Physics I — COR 212B	
11 AM			10:20 AM Advanced Mechanics		10:20 AM Advanced Mechanics	11:10 AM Astro Lab — COR 206	
Noon			Lunch		Lunch		
1 PM		12:20 PM Mathematical Methods of Theoretical		12:20 PM Mathematical Methods of Theoretical	Lab Preparation	1:10 PM Astro Lab — COR 206	
2 PM		Lunch		Lunch	2:00 PM Astro Group Meeting — SAMS 209		
3 PM				2:30 PM Mathematical Methods in Theoretical		Lunch	
4 PM					Office Hours		
5 PM		4:15 PM Communications in Physics — COR 104	Office Hours				
6 PM		Dinner	5:30 PM Cosmology — SAMS	Dinner	5:30 PM Cosmology — SAMS	Dinner	
7 PM		7:00 PM Homework	Dinner	7:00 PM Homework	7:10 PM Astro Lab	7:00 PM Homework	
8 PM			8:00 PM Homework				
9 PM							
10 PM					Dinner		
11 PM					10:00 PM Homework		